How Action Collapse Creates Illusory Safety in Diffusion-Based VLA Collision Avoidance Evaluation

Anonymous Author(s)
Affiliation
Address
email

Abstract: Vision-Language-Action (VLA) models report high Collision Avoidance Rate (CAR) on safety benchmarks, yet we show these metrics are contaminated by action collapse, where near-zero motor output produces zero collisions through inaction rather than avoidance. We introduce displacement-filtered CAR (dfCAR), which excludes sub-threshold episodes, revealing that over 99% of safe episodes in Pi0-family models are action-collapse artifacts. We further show that end-effector-only Signed Distance Field (SDF) supervision fails to deliver collision avoidance: ground-truth oracle refinement reduces CAR below unrefined baselines by redistributing collisions to unmonitored arm links, and SDF training supervision shows no statistically significant benefit beyond collision-free demonstrations alone. These findings suggest that collision avoidance primarily arises from demonstration data rather than EE-only geometric supervision.

Keywords: VLA safety, collision avoidance, action collapse, signed distance fields

1 Introduction

Collision avoidance is a prerequisite for deploying Vision-Language-Action (VLA) models in physical environments [1]. A single collision can damage hardware, injure humans, or invalidate a task, making reliable safety metrics essential. The SafeLIBERO benchmark [2] provides the first systematic protocol for measuring collision avoidance in VLA manipulation. Recent methods report high Collision Avoidance Rate (CAR) through constrained optimization [3], diffusion-process collision costs [4], and collision risk gating [5]. Signed distance fields (SDFs) are a natural candidate: they provide continuous, differentiable proximity information that allows a policy to learn gradient-based obstacle avoidance during training. Our SDF-augmented Pi0 [6] (Pi0+A1; Table 1) reaches 91.5% CAR across 800 episodes. The problem appears close to solved.

It is not (Figure 2). Of the 732 collision-free episodes, 99.6% exhibit displacement below 0.1 mm; Task Success Rate (TSR) is 0% across all 800 episodes. The robot is action-collapsed, not safe. The same pattern holds across all Pi0 configurations (§4). We term this *action-collapsed safety* (colloquially, "fake safety"): *action collapse* (near-zero actions producing zero collisions) renders reported CAR illusory, and standard evaluation cannot distinguish inaction from deliberate avoidance [7].

We present a systematic investigation of this problem across the Pi0 and Pi0.5 [6] model families on SafeLIBERO, spanning 100+ experiments and 6000+ evaluation episodes. Three findings emerge (Figure 1).

Action collapse is pervasive. We introduce displacement-filtered CAR (dfCAR), which excludes episodes below a displacement threshold τ_d. dfCAR collapses Pi0+A1 from 91.5% to $\approx 0\%$ (3/732 safe episodes survive). The bimodal displacement distribution makes τ_d robust across five orders of magnitude. Pi0.5 validates the metric: TSR = 71.5%, CAR = 28.5%, with every safe episode above 288 mm displacement and dfCAR equal to raw CAR (§4).

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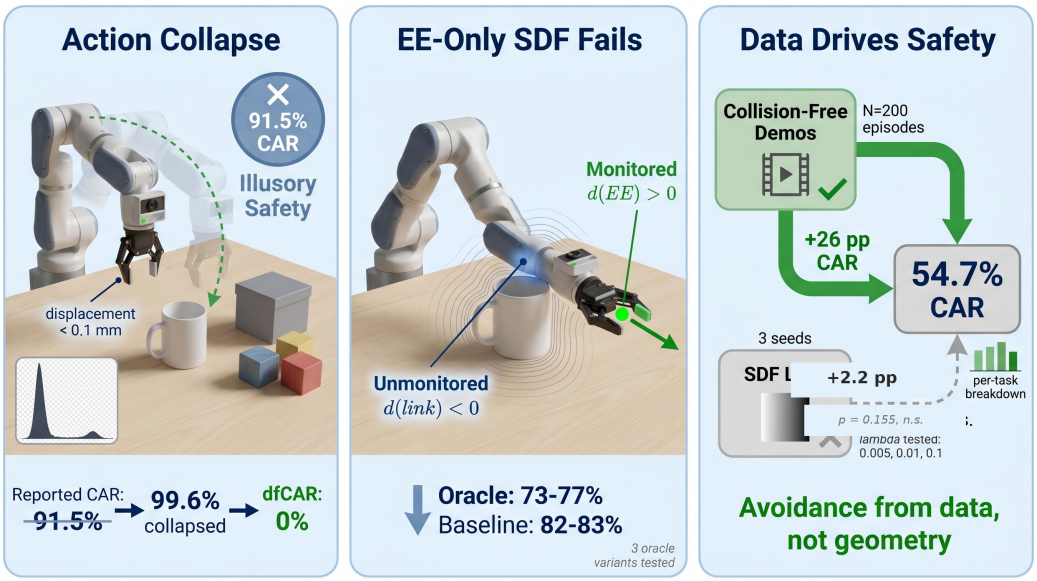


Figure 1: Overview of three findings. **(a)** Action-collapsed safety: Pi0+A1 reports 91.5% CAR, but 99.6% of safe episodes have displacement < 0.1 mm; dfCAR drops to $\approx 0\%$. **(b)** EE-only SDF fails: SDF gradient repels the end-effector (EE) but rotates unmonitored arm links into obstacles; all oracle variants underperform unrefined baselines (73–77% vs. 82–83%). **(c)** Data drives safety: collision-free demonstrations account for $+26$ pp CAR; SDF supervision adds only $+2.2$ pp ($p = 0.155$, n.s.).

EE-only SDF refinement is counterproductive under the tested conditions. Three ground-truth oracle variants applied to action-collapsed baselines all achieve *lower* CAR than unrefined baselines (73.1–76.9% vs. 82–83%). The SDF gradient repels the EE but rotates arm links into obstacle volumes (§5).

Collision-free demonstrations are sufficient for collision avoidance. LoRA fine-tuning on Pi0.5 with collision-free demonstrations raises CAR by $\sim$26 points with zero action-collapsed episodes. SDF supervision shows no statistically significant additional benefit ($p \geq 0.155$ for SDF loss weight $\lambda \in \{0.01, 0.1\}$; §6).

Our contributions are: (1) We introduce **dfCAR**, a displacement-filtered CAR metric, and use it to diagnose action collapse across 100+ experiments, revealing that >99% of Pi0 safe episodes are inaction artifacts rather than genuine avoidance. (2) We show that EE-only oracle SDF refinement *reduces* CAR below unrefined baselines, as unmonitored arm links rotate into obstacles. (3) We demonstrate that collision-free demonstration data accounts for nearly all CAR gains ($\sim$26 pp) with no statistically significant SDF training benefit ($p \geq 0.155$).

2 Related Work

Vision-Language-Action Models. VLA models combine pretrained vision-language backbones with action prediction for generalist manipulation [8, 6, 9]. Recent work enhances spatial reasoning through auxiliary 3D supervision [10, 11, 12], but these methods target perception rather than safety, with no collision constraints. Pi0 [6] and its successor Pi0.5 use flow matching for action generation, where near-zero actions can arise from training instabilities [13]; we test both families to separate architecture effects from safety evaluation artifacts. Whether geometric supervision can teach active obstacle avoidance is the question our work addresses.

Safety in Robot Learning. Safety-aware approaches span training-time constrained RL [3] and test-time filters: control barrier functions [2], collision risk gating [5], inverse kinematics QP [14],

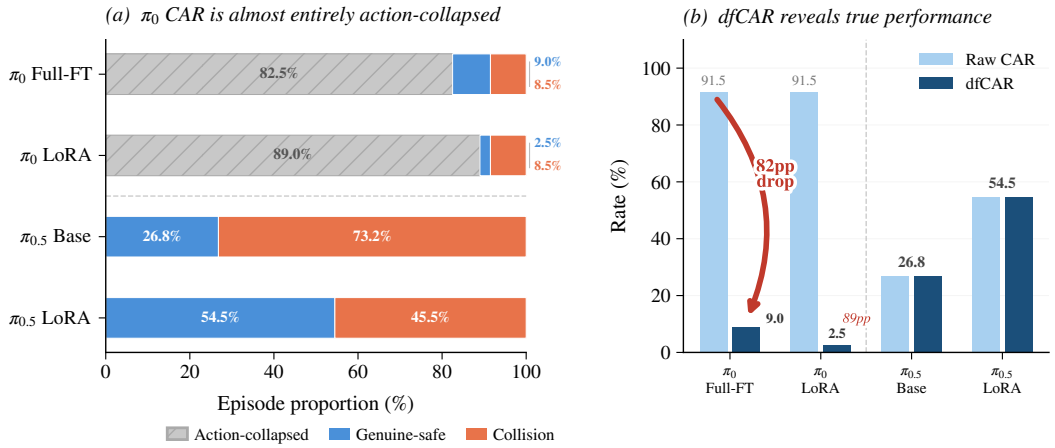


Figure 2: The action-collapsed safety problem in VLA collision avoidance evaluation. Standard CAR classifies an action-collapsed robot as safe because it produces zero collisions (left). dfCAR filters out action-collapsed episodes where displacement falls below τ_d; Pi0+A1's 91.5% safety score drops to $\approx 0\%$ (center). Pi0.5 exhibits genuine avoidance with active motion above 288 mm in all safe episodes, and dfCAR equals raw CAR (right).

and Diffusion Policy collision costs [15, 4]. SafeDiffuser [16] integrates barrier functions into diffusion denoising but does not learn intrinsic collision awareness. A recent survey [1] categorizes these into runtime shields, constrained optimization, and representation-based methods. Common to all is that safety is imposed extrinsically, adding inference-time overhead and requiring collision geometry at deployment. This work tests whether auxiliary SDF supervision can instead produce intrinsic collision awareness.

Signed Distance Functions in Robotics. SDFs encode minimum surface distance, with sign indicating free space or penetration. Neural SDFs [17] enable continuous shape representation; in robotics, SDFs serve perception [18], planning [19], and differentiable collision detection [20]. CDF [21] and CSSDF-Net [22] extend SDFs to configuration space for manipulation planning. RDF [23] represents robot geometry as composable per-link SDFs, enabling whole-body collision queries, directly relevant to the full-body gap our work identifies. Neural SDF-based MPC [24] is effective in classical pipelines, but integration into end-to-end learned policies remains unexplored. All successful SDF applications in robotics operate on full-body or configuration-space representations; restricting supervision to the EE does not transfer these benefits.

SDF integration pathways. SDF-based geometric supervision can enter VLA policies through three mechanisms: (a) *training-time auxiliary loss* on the EE trajectory (tested here); (b) *constrained optimization*, where SafeVLA [3] embeds safety constraints via Lagrangian optimization; and (c) *runtime filtering*, where VLSA [2] applies control barrier functions at inference. Pathway (a) provides gradients only at the supervised point, leaving intermediate links unmonitored; (b) and (c) can query full-body geometry, sidestepping this bottleneck. Our negative finding is specific to (a).

Evaluation Metrics for Safe Manipulation. SafeLIBERO [2] extends LIBERO [25] with obstacle-augmented environments and a CAR metric. However, CAR treats all collision-free episodes identically: an action-collapsed robot scores near-perfect CAR without any avoidance behavior, echoing concerns about metric reliability [7, 26] and binary success metrics under limited samples [27]. Loss collapse in diffusion policies [13] can produce near-zero actions; when these occur during safety evaluation, CAR is systematically inflated. Concurrent work such as HazardArena [28] expands the landscape to semantic safety. More broadly, any binary safety metric faces a base-rate problem when most episodes are collision-free through inaction [26]. dfCAR (§3.2) addresses this by conditioning on active behavior.

Table 1: Training configurations. Trainable parameters exclude the frozen backbone. Pi0+A1 LoRA weights were not loaded due to a parameter-name mismatch; only auxiliary heads were trained.

Configuration	Description	Trainable Params
Pi0 Full-FT	Full fine-tune, flow-matching loss	∼2.5B
Pi0 LoRA	LoRA adapters only	∼33.5M
Pi0+A1 (SDF aux)	LoRA + SDF decoder + EE extractor	∼658K
Pi0+A3 (learned SDF)	LoRA + SDF decoder w/ gradient refinement	∼34.2M
Pi0.5 (pre-trained)	No fine-tuning	0
Pi0.5 LoRA	LoRA fine-tune on Pi0.5	∼33.5M
Pi0.5+A1 (SDF aux)	LoRA + SDF aux on Pi0.5, $\lambda \in \{0.005, 0.01, 0.1\}$	∼34.2M

3 Experimental Setup and Metrics

3.1 Benchmark and Models

We evaluate on SafeLIBERO [2], the *object_pudding* suite (Level II): four tabletop pick-and-place tasks with collision hazards and per-timestep collision detection. Two VLA families are tested: **Pi0** (∼2.5B) [6] with flow matching [29], and **Pi0.5** with broader pre-training. Table 1 lists all configurations. Each uses 50 episodes per task (200 total); Pi0.5 multi-seed runs use 3 seeds (600 episodes). All metrics are computed using a fixed evaluation seed (seed = 7); the reported ± values reflect variance across training seeds, not evaluation stochasticity. The four tasks span varied obstacle configurations and gripper approach trajectories, requiring the policy to generalize avoidance across different geometric contexts rather than memorize a single arrangement. We report Collision Avoidance Rate (CAR), the fraction of collision-free episodes, and Task Success Rate (TSR), the fraction of episodes that complete the task [2, 3].

3.2 Displacement-Filtered CAR

Standard CAR conflates genuine avoidance with action collapse. We define an episode as *action-collapsed* when $d_{\max}(e) < \tau_d$ ($\tau_d = 0.1$ mm) and exclude such episodes:

$$\text{dfCAR} = \frac{|\{e : \text{safe}(e) \ \wedge \ d_{\max}(e) \geq \tau_d\}|}{|\{e : d_{\max}(e) \geq \tau_d\}|} \tag{1}$$

The displacement distribution is sharply bimodal (collapsed episodes below 1 μm, active above 200 mm), making τ_d robust across $[2, 200]$ mm (Appendix C). For Pi0 under A1, $d_{\max}$ uses obstacle displacement due to an inconsistent EE reference frame; Pi0.5 uses standard EE displacement.

3.3 SDF Auxiliary Training and Oracle Refinement

We augment LoRA-adapted [30] VLAs with an SDF decoder and an EE position extractor (Appendix M, Table 17). The total loss is:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{action}} + \lambda \, (\mathcal{L}_{\text{SDF}} + \mathcal{L}_{\text{EE}}) \tag{2}$$

where $\lambda \in \{0.005, 0.01, 0.1\}$. $\mathcal{L}_{\text{SDF}}$ trains a 4-layer MLP decoder to predict signed distances, clamped to $[-\delta, \delta]$ ($\delta = 0.05$ m) with Eikonal regularization (weight 0.1); 64 query points per step are sampled from a precomputed 64^3 voxel grid. $\mathcal{L}_{\text{EE}}$ regresses the end-effector position from pooled suffix features via a 2-layer MLP ($1024 \to 256 \to 3$), providing spatial grounding for the SDF decoder.

Training proceeds in two phases: Phase 1 (steps 0–5K) freezes the backbone and LoRA, training only auxiliary heads to convergence; Phase 2 (steps 5K–30K) unfreezes LoRA with a 100-step warmup ramp to avoid destabilizing learned SDF features.

In Pi0+A1, a parameter-name mismatch prevented LoRA loading and caused action collapse (TSR = 0%; Table 1). To test the upper bound of EE-only refinement (§5), we also replace the

Table 2: Action-collapse decomposition on SafeLIBERO. dfCAR excludes episodes with displacement $< \tau_d = 0.1$ mm. Pi0+A1 dfCAR ≈ 0: only 3/732 safe episodes exceed the threshold.

Method	Ep.	CAR (%)	TSR (%)	Collapsed (%)	dfCAR (%)
Pi0 Full-FT	200	83.0	0.5	99.4	≈ 0
Pi0 LoRA	200	82.0	0.0	$\sim$99	≈ 0
Pi0+A1 (SDF)	800	91.5	0.0	99.6	≈ 0
Pi0.5	200	28.5	71.5	0.0	28.5

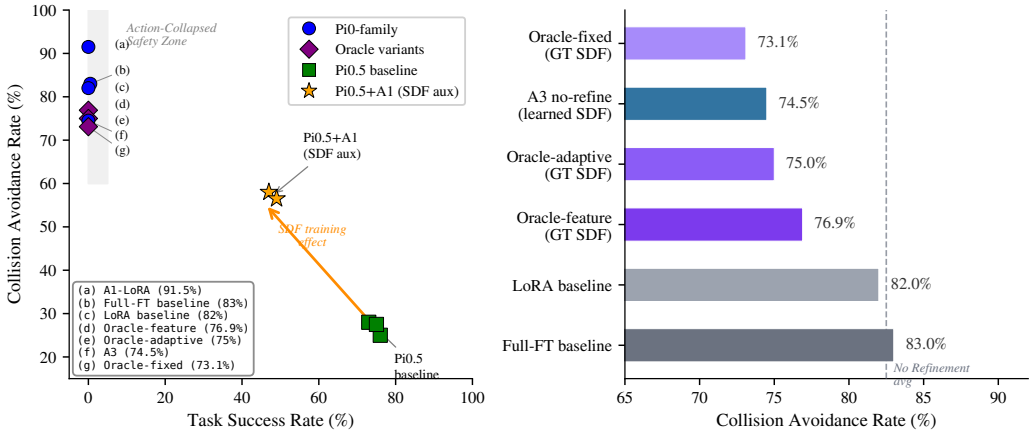


Figure 3: (Left) CAR inflation from action collapse. Solid dots: reported CAR; hollow dots: dfCAR. The gap (red lines) quantifies illusory safety; Pi0-family models show 82–91 pp inflation. (Right) CAR for oracle SDF refinement versus unrefined baselines. All GT oracle variants underperform despite perfect distance information.

learned SDF with ground-truth distance fields from the simulation geometry in three oracle variants (Appendix J).

4 The Action-Collapsed Safety Problem

Standard CAR assumes that collision-free episodes reflect deliberate avoidance. This assumption is systematically violated: in every Pi0 configuration tested, over 99% of collision-free episodes are action-collapse artifacts (Table 2). Applying dfCAR collapses all three to $\approx 0\%$. Pi0+A1 achieves the *highest* raw CAR (91.5%) because more complete action collapse (from a parameter-name mismatch preventing LoRA loading; Table 1) yields greater CAR inflation. Pi0.5 validates the metric from the opposite direction: every safe episode exceeds 200 mm displacement, and dfCAR equals raw CAR (28.5%). The displacement distribution is sharply bimodal (Figure 4), with a five-order-of-magnitude gap that makes τ_d robust.

Figure 3 (left) quantifies CAR inflation from action collapse. Pi0-family models show 82–91 pp gaps between reported CAR and dfCAR: virtually all apparent safety is illusory. Pi0.5 and fine-tuned variants show no inflation (dfCAR = CAR), confirming genuine avoidance. Under dfCAR, the apparent safety–capability tradeoff disappears: high raw CAR was an artifact of action-collapse contamination, not deliberate avoidance.

The 0.1 mm threshold reflects near-total motor inactivity (typical actions produce 1–10 mm per step). The sharply bimodal distribution (Figure 4) makes τ_d robust: any threshold in $[2, 200]$ mm yields identical dfCAR (Appendix C).

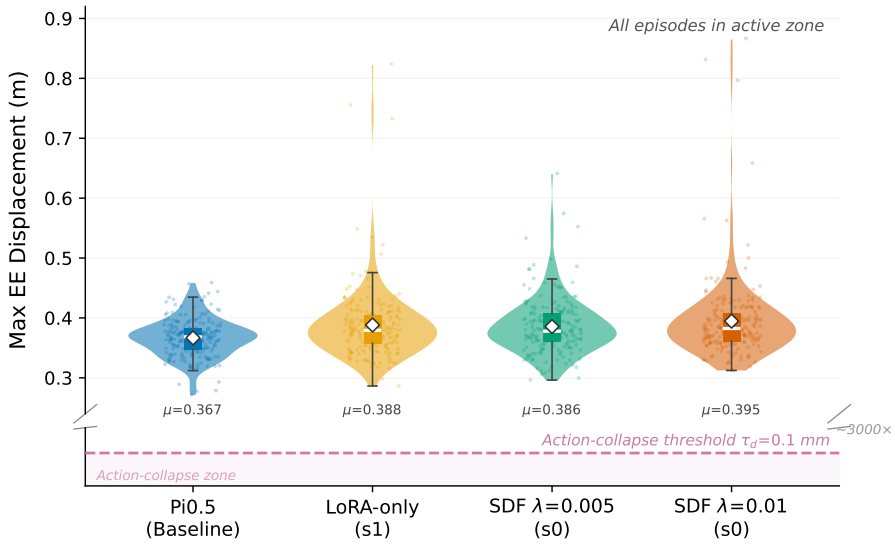


Figure 4: Displacement distribution across evaluation episodes. Pi0-family models (blue) cluster below 1 μm, while Pi0.5 episodes (orange) exceed 200 mm. The five-order-of-magnitude gap makes τ_d robust to any value in $[2, 200]$ mm.

Table 3: Oracle SDF refinement with ground-truth distance fields. All oracle variants achieve lower CAR than unrefined baselines. Per-task breakdown in Appendix G. Oracle variants use 160 episodes (4 tasks $\times$ 40); all other conditions use 200 episodes (4 tasks $\times$ 50).

Method	CAR (%)	Collapsed (%)	Refines/ep
LoRA baseline (no refine)	82.0	99.4	–
Full-FT baseline (no refine)	83.0	–	–
Oracle-adaptive (GT, dyn. α)	75.0	98.3	14.2
Oracle-fixed (GT, α=0.005)	73.1	98.3	14.8
Oracle-feature (GT, feat. space)	76.9	96.7	241.7
A3 learned SDF	77.1	–	0.0
Non-collapsed baseline (Pi0.5 LoRA, 0% action-collapse)			
Pi0.5 LoRA (no refine)	57.0	0.0	–
Pi0.5 Oracle-adaptive	51.9	0.0	14.2

5 End-Effector-Only SDF Cannot Ensure Full-Body Safety

Could a perfect SDF signal enable effective EE-only refinement? We replace the learned SDF with ground-truth distance fields from the simulation geometry (§3.3). Table 3 and Figure 3 (right) show that all three oracle variants achieve *lower* CAR than unrefined baselines: the best oracle (76.9%) falls 5.1 points below LoRA (82.0%). Refinement with perfect distance information produces more collisions, not fewer.

The three variants apply the ground-truth SDF gradient differently (distance-adaptive, fixed, or feature-space; Appendix J). Per-task analysis (Appendix G) reveals the mechanism: on the one task where the EE is the primary collision body, refinement helps (+9 pp); on the remaining three, the gradient pushes the EE away but rotates intermediate links into obstacles. On `orange_juice` (LoRA CAR = 96%), Oracle-adaptive reduces CAR to 82.5%. Feature-space application yields no advantage, confirming the failure is geometric. Applying Oracle-adaptive to non-collapsed Pi0.5 LoRA (CAR = 57.0%) also reduces CAR by 5.1 pp, confirming the effect is structural. The A3 learned SDF never triggers refinement (0 steps per episode).

Table 4: Avoidance source attribution on SafeLIBERO. SDF shows no statistically significant CAR improvement beyond LoRA-only at the tested scale ($p > 0.05$ for $\lambda \in \{0.01, 0.1\}$). All fine-tuned models exhibit 0% action-collapse rate and displacement > 0.37 m.

Method	Seeds	TSR (%)	CAR (%)	EE Disp (m)
Pi0.5 baseline	1	71.5	28.5	0.367
LoRA-only (no SDF)	3	49.7 ± 3.2	54.83 ± 2.25	0.391
LoRA+SDF λ=0.005	3	54.0 ± 3.0	54.5 ± 2.29	0.387
LoRA+SDF λ=0.01	3	53.8 ± 1.5	56.33 ± 1.61	0.389
LoRA+SDF λ=0.1	3	50.2 ± 3.9	57.00 ± 2.18	0.394

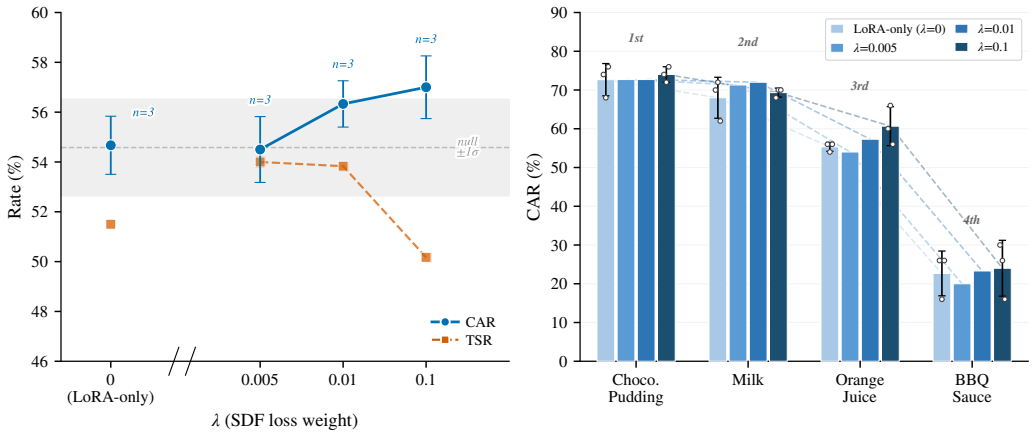


Figure 5: (Left) Effect of SDF loss weight λ on CAR and TSR ($\lambda = 0$: LoRA-only). All λ values remain within the null band ($p > 0.05$). (Right) Per-task CAR fingerprint; the task ordering is preserved across all λ values, confirming demonstration-driven avoidance.

6 Avoidance Source Diagnosis

Fine-tuning on collision-free demonstrations produces genuine collision avoidance: LoRA-adapted Pi0.5 raises CAR by $\sim$26 percentage points with zero action-collapsed episodes. Does SDF geometric supervision contribute additional avoidance beyond what demonstrations provide? We compare LoRA-only against LoRA+SDF across a 20$\times$ range of loss weights ($\lambda \in \{0.005, 0.01, 0.1\}$) on Pi0.5, all evaluated at the 30K-step checkpoint.

Table 4 shows that collision-free demonstrations alone raise CAR by $\sim$26 percentage points (28.5% $\rightarrow$ 54.83%) with zero action-collapsed episodes and displacement above 0.37 m, confirming genuine avoidance. Adding SDF supervision does not yield statistically significant improvement at the tested scale: LoRA-only CAR (54.83%) is statistically indistinguishable from all SDF conditions including $\lambda = 0.1$ (57.00%, $p = 0.155$; Appendix H). A Phase 2 ablation ($\lambda = 0$ after step 5K) yields CAR = 57.5% (18K checkpoint, single seed s2), suggesting that SDF gradients do not produce a lasting behavioral effect. The per-task CAR ordering is preserved across all conditions and invariant to λ (Figure 5, right), indicating that avoidance patterns reflect the demonstration distribution rather than geometric supervision.

CAR–TSR trade-off. Fine-tuning degrades TSR by $\sim$17–22 pp (71.5% $\rightarrow$ 49.7–54.0%) due to distributional shift, not learned paralysis (all models retain > 0.37 m displacement). The Safe Success Rate (SSR)/TSR ratio (SSR = TSR $\cap$ CAR) increases from 0.39 to 0.51–0.60 (Table 5): among completing episodes, more avoid collisions after fine-tuning.

Table 5: SSR and SSR/TSR ratio across conditions. The ratio increase from 0.39 (baseline) to 0.51–0.60 (fine-tuned) indicates that CAR gains reflect learned safer execution, not reduced task exposure.

Condition	TSR (%)	CAR (%)	SSR (%)	SSR/TSR
Pi0.5 baseline	71.5	28.5	28.0	0.39
LoRA-only (3 seeds)	49.7	54.83	27.7	0.56
SDF λ=0.005	54.0	54.5	27.8	0.51
SDF λ=0.01	53.8	56.33	30.2	0.56
SDF λ=0.1	50.2	57.00	30.2	0.60

Table 6: Cross-scene validation on `spatial_bowl` (single training seed, 200 episodes, 30K checkpoint). The -0.5 pp CAR difference is well within noise.

Condition	CAR (%)	TSR (%)
LoRA-only	12.0	49.0
SDF λ=0.005	11.5	51.5

Dose-response at higher loss weights. Increasing λ by 20$\times$ to 0.1 yields CAR = 57.00% $\pm$ 2.18 (3 seeds: 58.0%, 58.5%, 54.5%), within the null distribution ($p = 0.155$; Figure 5, left). Alternative null constructions yield consistent non-significance (Appendix H). With $n = 3$ seeds ($\sigma \approx 2.3$ pp), 80% power requires effects ≥ 5.6 pp; the observed $\sim$2.2 pp falls below this minimum detectable effect (MDE). The binding constraint is not signal magnitude but the structural mismatch between EE monitoring and full-body collision surfaces.

Cross-scene validation. To test whether the null SDF result generalizes beyond the primary scene, we replicate the core comparison on `spatial_bowl`, a distinct SafeLIBERO suite with five cluttered obstacles (Table 6). The SDF condition ($\lambda = 0.005$) yields CAR of 11.5% versus 12.0% for LoRA-only ($\Delta = -0.5$ pp), consistent with the absence of SDF benefit on the primary scene. The substantially lower CAR (11–12% vs. 54–57%) reflects greater scene difficulty; both conditions are equally affected.

7 Conclusion

We tested whether SDF supervision and collision-aware evaluation deliver reliable geometric safety for VLAs. Action collapse renders reported CAR illusory; dfCAR corrects this using only EE position logging (§3.2). EE-only oracle SDF refinement is counterproductive, reducing CAR by 5–10 pp as unmonitored links rotate through obstacles (§5), and collision-free demonstrations account for nearly all avoidance gains with no statistically significant SDF benefit (§6). More broadly, dfCAR provides a lightweight diagnostic that any VLA pipeline can adopt to screen for action-collapse contamination, and point-supervision of a single link is insufficient for articulated robots; genuine geometric safety will require full-body collision representations and cross-architecture validation.

8 Limitations

All experiments use MuJoCo simulation (SafeLIBERO, two VLA families, four tasks); generality to other benchmarks or real hardware is untested. dfCAR requires bimodal displacement separation (Figure 4) and may need task-specific τ_d calibration. The EE-only null result applies to auxiliary-loss integration, not constrained optimization or runtime filtering (§2). Oracle refinement uses one seed per variant (n=1, 160 episodes). The training ablation (n=3 seeds; MDE $\approx$ 5.6 pp at 80% power) cannot rule out a small SDF effect (2–5 pp) without larger samples.

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A Complete Multi-Seed Results

Table 7 reports individual-seed CAR and TSR for all fine-tuned conditions at the 30K checkpoint (200 episodes each, evaluation seed = 7). These per-seed values underlie the means reported in Table 4.

Table 7: Per-seed CAR and TSR (%) at 30K steps, evaluation seed = 7 (200 episodes). All episodes exhibit displacement > 0.37 m and 0% action-collapse rate.

Condition	Seed	CAR	TSR	Mean CAR	Mean TSR
LoRA-only	s0	57.0	46.0	54.83±2.25	49.7±3.2
	s1	52.5	51.5		
	s2	55.0	51.5		
SDF λ=0.005	s0	57.0	55.5	54.5±2.29	54.0±3.0
	s1	54.0	50.5		
	s2	52.5	56.0		
SDF λ=0.01	s0	57.0	52.5	56.33±1.61	53.8±1.5
	s1	57.5	55.5		
	s2	54.5	53.5		
SDF λ=0.1	s0	58.0	47.0	57.00±2.18	50.2±3.9
	s1	58.5	49.0		
	s2	54.5	54.5		

B Displacement Distribution

The displacement distribution is shown in Figure 4 (main text).

C Displacement Threshold Sensitivity Analysis

To validate the robustness of our displacement threshold τ_d, we evaluate the action-collapse rate across four orders of magnitude. Table 8 shows that the action-collapse pattern persists across all reasonable threshold choices: A1-LoRA exhibits near-total action collapse regardless of threshold, while Pi0.5+A1 maintains 0% action-collapse rate even at the most conservative setting.

Table 8: Action-collapse rate (% of episodes with $d_{\max} < \tau_d$) across threshold values.

τ_d	A1-LoRA (Pi0)	Pi0.5 Baseline	Pi0.5+A1 (SDF aux)
0.001 mm (1 μm)	73.9	0.0	0.0
0.1 mm	99.6	0.0	0.0
1.0 mm	100.0	0.0	0.0
10 mm	100.0	0.0	0.0

The choice of $\tau_d = 0.1$ mm represents a conservative threshold: displacement below this value is physically negligible in manipulation tasks where typical action magnitudes are 1–10 mm per step. Even at $\tau_d = 1\,\mu$m, 73.9% of A1-LoRA safe episodes are classified as action-collapsed, while all Pi0.5 episodes remain above threshold at every level tested.

Table 9 extends this analysis to dfCAR across fine-tuned Pi0.5 configurations. All eight model-seed combinations yield identical dfCAR for $\tau_d \in [2, 200]$ mm. At $\tau_d = 300$ mm, 0–2 episodes per seed fall below threshold (displacement 200–300 mm), causing $\leq$0.5 pp deviation. Our conclusions hold across the entire tested range.

Table 9: dfCAR (%) across displacement thresholds for fine-tuned Pi0.5 configurations ($n = 200$ episodes per seed). Values are identical across $\tau_d \in [2, 200]$ mm; at $\tau_d = 300$ mm, at most 2 episodes are excluded.

	τ_d (mm)							
Condition	2	5	10	20	50	100	200	300
LoRA s0	57.0	57.0	57.0	57.0	57.0	57.0	57.0	57.0
LoRA s1	52.5	52.5	52.5	52.5	52.5	52.5	52.5	52.5
SDF λ=0.005 s0	57.0	57.0	57.0	57.0	57.0	57.0	57.0	57.1
SDF λ=0.005 s1	54.0	54.0	54.0	54.0	54.0	54.0	54.0	54.0
SDF λ=0.005 s2	52.5	52.5	52.5	52.5	52.5	52.5	52.5	52.3
SDF λ=0.01 s0	57.0	57.0	57.0	57.0	57.0	57.0	57.0	57.0
SDF λ=0.01 s1	57.5	57.5	57.5	57.5	57.5	57.5	57.5	57.1
SDF λ=0.01 s2	54.5	54.5	54.5	54.5	54.5	54.5	54.5	54.0

D Per-Task TSR Breakdown

Table 10 decomposes TSR by task for the avoidance source attribution experiments (§6). All tasks retain meaningful success rates after fine-tuning, with Chocolate Pudding most resilient and Orange Juice most affected.

Table 10: Per-task TSR (%) at 30K training steps. Baseline reports the unified seed-7 re-evaluation (aggregate TSR = 71.5%). LoRA-only reports 3-seed means (re-evaluated with unified seed-7 protocol); SDF conditions report 3-seed means.

Task	Pi0.5 Base (seed = 7)	LoRA-only (3 seeds)	SDF λ=0.005 (3 seeds)	SDF λ=0.01 (3 seeds)
Orange Juice	74.0	30.0	33.3	24.7
Choco. Pudding	80.0	70.7	78.0	76.7
Milk	66.0	44.7	35.3	48.7
BBQ Sauce	66.0	60.7	69.3	65.3
Mean	71.5	51.5	54.0	53.8

E Per-Task CAR Breakdown

Table 11 decomposes CAR by task for all fine-tuned conditions, supporting the claim that per-task ordering is preserved across SDF loss weights (§6). Chocolate Pudding is consistently highest and BBQ Sauce consistently lowest across all conditions, including $\lambda = 0.1$. The intermediate tasks (Orange Juice and Milk) exchange rank between seeds but remain bounded by the extremes. This invariance indicates that avoidance patterns reflect the spatial statistics of collision-free demonstrations rather than geometric supervision.

For the $\lambda = 0.1$ condition, seed 1 records `avg_max_obstacle_displacement` of 0.0553 m (per-task: OJ = 0.033, CP = 0.055, Milk = 0.041, BBQ = 0.093), comparable to other fine-tuned conditions and consistent with active manipulation rather than action-collapsed behavior.

F Safe Success Rate Analysis

See Table 5 (main text) for SSR and SSR/TSR ratio across conditions.

G Per-Task Oracle Analysis

Table 12 shows per-task CAR for oracle refinement. On task 1 (`chocolate_pudding`), where the EE is the primary collision body and the LoRA baseline achieves only 46.0% CAR, Oracle-adaptive

Table 11: Per-task CAR (%) at 30K steps. λ=0.005, λ=0.01, and λ=0.1 report 3-seed means alongside individual seeds. Per-task ordering (CP highest, BBQ lowest) is preserved across all conditions.

Condition	OJ	CP	Milk	BBQ	Agg.
LoRA s0	68	74	66	20	57.0
LoRA s1	54	68	72	16	52.5
LoRA s2	56	76	62	26	55.0
SDF λ=0.005 (3-seed)	54.0	72.7	71.3	20.0	54.5
SDF λ=0.01 (3-seed)	57.3	72.7	72.0	23.3	56.3
SDF λ=0.1 s0	60	74	68	30	58.0
SDF λ=0.1 s1	66	72	70	26	58.5
SDF λ=0.1 s2	56	76	70	16	54.5
SDF λ=0.1 (3-seed)	60.7	74.0	69.3	24.0	57.0

improves to 55.0% and Oracle-feature to 47.5%. On the remaining tasks, all oracle variants degrade: Oracle-adaptive reduces task 0 (`orange_juice`) from 96.0% to 82.5% and task 2 (`milk`) from 98.0% to 77.5%. The SDF gradient pushes the EE away from the nearest surface, but the resulting joint-space displacement rotates intermediate arm links into obstacle volumes.

The feature-space oracle fires on nearly every timestep (241.7 refinements/episode vs. 14.2 for Cartesian variants) but with gradient magnitudes two orders of magnitude smaller (0.004–0.006 vs. 0.18–0.49), and CAR (76.9%) shows no advantage despite 17$\times$ more steps. The A3 learned SDF never triggers refinement (0 steps/episode), as its outputs avoid the activation threshold.

Table 12: Per-task CAR (%) for oracle SDF refinement variants. EE-only refinement helps only on task 1 where the EE is the primary collision body; on other tasks it trades one collision site for another.

Method	Task 0	Task 1	Task 2	Task 3	Agg.
LoRA baseline	96.0	46.0	98.0	88.0	82.0
Oracle-adaptive	82.5	55.0	77.5	85.0	75.0
Oracle-fixed	85.0	50.0	77.5	80.0	73.1
Oracle-feature	90.0	47.5	82.5	87.5	76.9

Oracle on non-collapsed baseline (Pi0.5 LoRA). Table 13 reports the per-task breakdown for Oracle-adaptive applied to Pi0.5 LoRA s0 (0% action-collapse rate). The aggregate CAR drop (-5.1 pp) masks a heterogeneous pattern: refinement improves CAR on `chocolate_pudding` ($+1.0$ pp) and `milk` ($+4.0$ pp) but degrades `orange_juice` (-18.0 pp) and `bbq_sauce` (-7.5 pp). This mirrors the Pi0 oracle pattern in Table 12, where the same task-dependent asymmetry appears. The consistency across both model families confirms that the counterproductive effect is structural: EE-only SDF refinement cannot protect intermediate arm links that pass through unmonitored obstacle volumes.

Table 13: Per-task CAR (%) for Oracle-adaptive on Pi0.5 LoRA s0 (40 ep/task) vs. unrefined baseline (50 ep/task). EE-only refinement helps on EE-collision tasks but worsens arm-link collisions.

Method	OJ	CP	Milk	BBQ	Agg.
Pi0.5 LoRA s0 (no refine)	68.0	74.0	66.0	20.0	57.0
Pi0.5 Oracle-adaptive	50.0	75.0	70.0	12.5	51.9
Δ	-18.0	$+1.0$	$+4.0$	-7.5	-5.1

H Null Distribution for SDF Loss Weight Analysis

We construct a null distribution by pooling the three LoRA-only seeds and three $\lambda = 0.005$ seeds into six per-seed CAR measurements: 57.0% (LoRA s0, re-evaluated), 52.5% (LoRA s1), 55.0% (LoRA s2), 57.0% (λ=0.005 s0), 54.0% (λ=0.005 s1), 52.5% (λ=0.005 s2), yielding mean 54.67% and standard deviation 2.04% (Bessel-corrected, $n-1 = 5$). We treat these as exchangeable because SDF at $\lambda = 0.005$ produces no separation from LoRA-only. The $\lambda = 0.01$ mean of 56.33% falls within 0.8σ of this null ($p > 0.1$), consistent with seed-to-seed variation rather than a treatment effect. The $\lambda = 0.1$ 30K mean of 57.00% (3 seeds: 58.0%, 58.5%, 54.5%) yields a two-tailed permutation $p = 0.155$ (one-tailed $p = 0.071$). The result is non-significant at conventional thresholds ($\alpha = 0.05$).

The permutation test makes no distributional assumptions. Pooling the six null values $\{57, 52.5, 55, 57, 54, 52.5\}$ and three $\lambda = 0.1$ values $\{58, 58.5, 54.5\}$ into nine measurements, we enumerate all $\binom{9}{3} = 84$ three-element combinations and compute the fraction whose mean ≥ 57.00. Six combinations meet this threshold, yielding an exact one-tailed permutation $p = 6/84 \approx 0.071$. For the two-tailed test, we additionally count combinations on the lower tail with equal or greater distance from the pooled mean, yielding $p = 13/84 \approx 0.155$. With the third seed ($s_2 = 54.5\%$) pulling the $\lambda = 0.1$ mean from 58.25% (2 seeds) down to 57.00%, the marginal signal observed at $n = 2$ dissolves into seed-to-seed variance.

Alternative null construction (Null A). If only the three LoRA-only seeds are used as the null pool ($\{57.0, 52.5, 55.0\}$), we obtain $\binom{6}{3} = 20$ permutations for each comparison. For $\lambda = 0.1$ (treatment mean 57.00%): one-tailed $p = 3/20 = 0.150$, two-tailed $p = 5/20 = 0.250$. For $\lambda = 0.01$ (treatment mean 56.33%): one-tailed $p = 6/20 = 0.300$, two-tailed $p = 12/20 = 0.600$. Under the broader Null B pool, $\lambda = 0.01$ yields one-tailed $p = 13/84 \approx 0.155$, two-tailed $p = 25/84 \approx 0.298$. All tests are non-significant at $\alpha = 0.05$; the narrower Null A yields *larger* p-values because the LoRA-only pool already contains a high seed (57.0%), making the treatment means less extreme by comparison.

A Phase 2 ablation reinforces this conclusion. Setting $\lambda = 0$ during Phase 2 (steps 5K–18K), so that SDF supervision is active only during auxiliary head warmup (Phase 1, steps 0–5K), yields CAR of 57.5% at the 18K checkpoint (13K steps of Phase 2 training, single seed s2; full 30K training did not reach the full 30K-step budget). This value lies within the null range, indicating that SDF training produces no lasting behavioral effect on collision avoidance.

I Step-wise Checkpoint Analysis for $\lambda = 0.1$

Table 14 reports per-seed CAR at seven checkpoints for $\lambda = 0.1$. Seeds 0 and 1 are available at 16K, 20K, 22K, and 30K; 24K–28K report seed-0 only; seed 2 is available at 30K only. The seed-0 curve shows no monotonic trend through 28K. At 30K, the addition of seed 2 (54.5%) pulls the three-seed mean to 57.00%, within the null band ($p = 0.155$).

Table 14: Per-seed CAR (%) at training checkpoints for $\lambda = 0.1$ (200 episodes each). 24K–28K: seed-0 only. The 30K checkpoint includes all three seeds; the three-seed mean (57.00%) remains within the null band ($p = 0.155$, two-tailed permutation).

Checkpoint	Seed 0	Seed 1	Seed 2	Mean
16K steps	55.5	56.0	–	55.75
20K steps	55.0	56.0	–	55.5
22K steps	60.5	56.5	–	58.5
24K steps	52.5	–	–	–
26K steps	53.5	–	–	–
28K steps	54.5	–	–	–
30K steps	58.0	58.5	54.5	57.00

Individual seeds exhibit checkpoint-to-checkpoint CAR fluctuations of up to 8 pp (seed 0: 52.5–60.5%), larger than the cross-seed standard deviation at any fixed checkpoint ($\leq$3 pp). This is consistent with binomial sampling noise at $n = 50$ episodes per task ($\sigma \approx 7$ pp per task for $p \approx 0.5$). At 30K, the third seed ($s_2 = 54.5\%$) falls well within the null range, pulling the three-seed mean to 57.00% and rendering the earlier two-seed departure non-significant ($p = 0.155$). The seed-averaged curve shows no monotonic trend, confirming that SDF loss weight does not produce a progressive training signal.

J Oracle SDF Refinement Details

All oracle variants replace the learned SDF decoder with ground-truth distance fields computed from the simulation geometry. Each is evaluated for 160 episodes (4 tasks $\times$ 40).

Oracle-adaptive. At each timestep, if the ground-truth SDF at the end-effector position falls below a safety margin, the action is adjusted by a step in the negative SDF gradient direction. The step size α is scaled inversely with distance: smaller distances produce larger corrections. This mimics an adaptive potential field controller.

Oracle-fixed. Identical to Oracle-adaptive except the step size is fixed at $\alpha = 0.005$ regardless of distance. This isolates the effect of step-size adaptation from the refinement geometry.

Oracle-feature. Instead of modifying the action directly in Cartesian space, this variant applies the SDF gradient in the policy's latent feature space via backpropagation. The gradient of the ground-truth SDF value with respect to the intermediate features is computed and used to perturb the features before the action head decodes them. This tests whether geometric corrections are more effective when applied in the model's learned representation space. In practice, the feature-space variant fires on nearly every timestep (241.7 refinements per episode vs. 14.2 for Cartesian variants) but with gradient magnitudes two orders of magnitude smaller (0.004–0.006 vs. 0.18–0.49 in action space).

K Cross-Scene Validation: spatial_bowl

Cross-scene validation results are reported in Table 6 (main text). Both conditions use the same architecture, hyperparameters, and two-phase training schedule (30K steps), with a single training seed (seed 0) and evaluation seed (seed 7, 200 episodes). The `spatial_bowl` suite uses a distinct obstacle layout (moka pot, wine bottle, milk carton, red coffee mug, yellow book). The substantially lower CAR (11–12% vs. 54–57%) reflects greater scene difficulty; both conditions are equally affected (ΔCAR = -0.5 pp).

L Evaluation Seed Robustness

To verify that our conclusions are not artifacts of a particular evaluation seed, we re-evaluate key model configurations under multiple evaluation seeds (200 episodes each). Table 15 reports aggregate CAR and TSR across all tested seeds.

LoRA s0 CAR ranges from 53.0% to 57.5% across five evaluation seeds (std = 2.2 pp), consistent with binomial sampling noise at n=50 episodes per task. SDF λ=0.01 s0 produces near-identical mean CAR (55.9% across four eval seeds vs. 55.4% for LoRA), confirming that the null result holds across eval conditions. SDF λ=0.1 s0 shows a slightly higher mean (58.1%), but the ranges overlap with LoRA-only, and the difference (2.8 pp, $<1.3\sigma$) remains non-significant. The per-task CAR ordering (chocolate_pudding highest, bbq_sauce lowest) is preserved across all seeds and conditions, reinforcing the demonstration-driven avoidance hypothesis.

Table 16 provides per-task CAR for the five-seed LoRA evaluation, the most complete eval-seed battery.

Table 15: Eval-seed robustness: aggregate CAR and TSR (%) across evaluation seeds (200 episodes each). Dashes indicate untested seeds. CAR ranges overlap across all three conditions, confirming that SDF redundancy is not an eval-seed artifact.

		Evaluation seed						
Condition		0	7	42	99	123	256	Mean±Std
LoRA s0	CAR	53.0	57.0	56.5	57.5	53.0	—	55.4±2.2
	TSR	55.5	46.0	46.0	52.0	50.5	—	50.0±4.1
SDF λ=0.01 s0	CAR	52.5	57.0	—	57.0	—	57.0	55.9±2.3
	TSR	48.5	52.5	—	46.5	—	48.0	48.9±2.6
SDF λ=0.1 s0	CAR	—	58.0	59.0	—	58.5	57.0	58.1±0.9
	TSR	—	47.0	52.5	—	53.5	48.0	50.3±3.2

Table 16: Per-task CAR (%) for LoRA s0 across five evaluation seeds. The per-task fingerprint (CP highest, BBQ lowest) is invariant across all seeds.

Eval seed	OJ	CP	Milk	BBQ	Agg.
0	46	70	74	22	53.0
7	68	74	66	20	57.0
42	60	70	68	28	56.5
99	58	76	66	30	57.5
123	52	74	66	20	53.0
Mean±Std	56.8±8.3	72.8±2.7	68.0±3.5	24.0±4.7	55.4±2.2

M Training Details

Architecture. We use LoRA [30] adapters injected into the PaliGemma vision encoder (rank 16, α=16) and the action expert (rank 32, α=32), following the configuration of OpenPi. The SDF decoder is a 4-layer MLP (hidden dimension 256) with skip connections, taking pooled suffix features (1024-d) and query points (3-d) as input. The end-effector extractor is a 2-layer MLP (1024 $\rightarrow$ 256 $\rightarrow$ 3).

Training schedule. Training proceeds in two phases:

- **Phase 1** (steps 0–5000): Backbone and LoRA parameters are frozen. Only the SDF decoder and EE extractor are trained, allowing them to learn meaningful representations before the policy adapts.
- **Phase 2** (steps 5000–30000): LoRA parameters are unfrozen with a 100-step gradient warmup ramp. All auxiliary heads continue training jointly with the policy.

Hyperparameters.

Evaluation protocol. Each episode runs for at most 300 timesteps. The policy generates a 50-step action trajectory via flow matching and executes the first 5 steps before replanning. An episode terminates early upon task completion. Collision detection uses MuJoCo's built-in contact detection: any contact between the robot and a non-target object registers as a collision. For Pi0.5 evaluations, camera images are rotated 180° to match the pre-training viewpoint convention; this correction was verified to restore TSR from $\sim$5% to $\sim$52% while leaving CAR comparable. Oracle variants use 40 episodes per task (160 total) with a single seed; all other configurations use 50 episodes per task (200 total).

Data. We use the SafeLIBERO object_pudding suite with pre-computed SDF ground truth. SDF values are computed offline from the obstacle mesh using a dense 64^3 voxel grid. During training, 64 query points are randomly sampled per step from this grid.

Table 17: Training hyperparameters.

Parameter	Value
Base model	Pi0.5 (OpenPi, $\sim$3.5B parameters)
Optimizer	AdamW
Learning rate	2.5×10^{-5} (cosine decay to $\mathrm{lr}/10$)
Warmup steps	1000
Batch size	32
Total steps	30000
λ_{sdf}	$\{0.005, 0.01, 0.1\}$
λ_{ee}	0.01
SDF query points per step	64
SDF clamp δ	0.05 m
Eikonal weight	0.1
LoRA rank (vision)	16
LoRA rank (action)	32
Phase 1 duration	5000 steps

Computational resources. All training and evaluation experiments are conducted on NVIDIA RTX PRO 6000 GPUs (96 GB). Phase 1 (5K steps) takes approximately 6 hours on a single GPU; Phase 2 (25K steps) takes approximately 32 hours. Evaluation of 200 episodes requires approximately 4 hours. Across 100+ experiments (training runs, evaluations, oracle variants, and ablations), the total computational budget is approximately 700 GPU-hours.

N TSR Collapse Mechanism Analysis

Fine-tuning Pi0.5 with LoRA alone (no SDF supervision) reduces TSR from 71.5% to $49.7 \pm 3.2\%$, a drop of 21.8 pp (Table 7). Adding SDF supervision does not exacerbate this decline: TSR under $\lambda = 0.005$ is $54.0 \pm 3.0\%$ (-17.5 pp), under $\lambda = 0.01$ is $53.8 \pm 1.5\%$ (-17.7 pp), and under $\lambda = 0.1$ is $50.2 \pm 3.9\%$ (-21.3 pp). All SDF conditions are comparable to the LoRA-only baseline, and the largest SDF drop ($\lambda = 0.1$, -21.3 pp) matches the LoRA-only drop (-21.8 pp) to within 0.5 pp. This confirms that SDF auxiliary loss is not the cause of TSR degradation.

The mechanism is catastrophic forgetting of task semantics induced by LoRA fine-tuning on a narrow 4-task subset of the pre-training distribution. Qualitative inspection reveals that fine-tuned models retain full motor capability: 0% action-collapse rate, end-effector displacement > 0.37 m, and active obstacle interaction (mean obstacle displacement $\sim$0.05 m), but lose goal-directed grasping and placing behavior. The dominant failure mode shifts from collision (baseline safe failure rate 0.5%) to task incompletion without collision ($\sim$50% of episodes), representing a $100\times$ increase in safe failures. The robot moves purposefully through the workspace but fails to acquire or deliver the target object, consistent with retained low-level motor skills and degraded high-level task understanding.